

28.1

Numerical Methods (6)

28.1.1 Numerical Methods: NIELIT 2016 MAR Scientist B - Section B: 7



In which of the following methods proper choice of initial value is very important?

- A. Bisection method
C. Newton-Raphson
B. False position
D. Bairsto method

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Answer key

28.1.2 Numerical Methods: NIELIT 2016 MAR Scientist C - Section B: 1



Choose the most appropriate option.

The Newton-Raphson iteration $x_{n+1} = \frac{x_n}{2} + \frac{3}{2x_n}$ can be used to solve the equation

- A. $x^2 = 3$ B. $x^3 = 3$ C. $x^2 = 2$ D. $x^3 = 2$

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28.1.3 Numerical Methods: NIELIT 2017 DEC Scientist B - Section B: 3



Using bisection method, one root of $x^4 - x - 1$ lies between 1 and 2. After second iteration the root may lie in interval:

- A. (1.25, 1.5) B. (1, 1.25)
C. (1, 1.5) D. None of the options.

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Answer key

28.1.4 Numerical Methods: NIELIT 2017 OCT Scientific Assistant A (CS) - Section C: 7



The convergence of the bisection method is

- A. Cubic B. Quadratic C. Linear D. None

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Answer key

28.1.5 Numerical Methods: NIELIT 2021 Dec Scientist A - Section B: 89



The real root of the equation $x^3 - x - 5 = 0$ lying between 1 and 2 after first iteration by Netwon-Raphson method is _____, if initial approximation is taken as $x_0 = 2 \in [1, 2]$:

A. 1.909

B. 1.904

C. 1.921

D. 1.940

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28.1.6 Numerical Methods: NIELIT 2021 Dec Scientist B - Section B: 60

One root of $x^3 - x - 4 = 0$ lies in $(1, 2)$. In bisection method, after first iteration the root lies in the interval _____.

A. $(1, 1.5)$ B. $(1.5, 2)$ C. $(1.25, 1.75)$ D. $(1.75, 2)$

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28.2**Root Finding (1)****28.2.1 Root Finding: NIELIT 2021 Dec Scientist B - Section B: 106**

One root of $x^3 - x - 4 = 0$ lies in $(1, 2)$. In bisection method, after first iteration the root lies in the interval _____.

A. $(1, 1.5)$ B. $(1.5, 2)$ C. $(1.25, 1.75)$ D. $(1.75, 2)$

nielit2021dec-scientistb numerical-methods root-finding analytical-aptitude

Answer key **28.3****Vector Space (1)****28.3.1 Vector Space: NIELIT 2017 DEC Scientist B - Section B: 20**

Let u and v be two vectors in R^2 whose Euclidean norms satisfy $|u| = 2|v|$. What is the value α such that $w = u + \alpha v$ bisects the angle between u and v ?

A. 2

B. 1

C. $\frac{1}{2}$

D. -2

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Answer key **Answer Keys**

28.1.1	C
28.1.6	B

28.1.2	A
28.2.1	B

28.1.3	B
28.3.1	A

28.1.4	C
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28.1.5	A
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